MULTI-DAY WATER TANK SIMULATION

Assignment Matter • Problem Statement • Algorithm • Pseudocode • Output

1. Problem Statement

Develop a Python program to simulate the water level in a tank over a given number of days. For each day, the program accepts the amount of water entering the tank (inflow) and leaving the tank (outflow), updates the water level, detects overflow or shortage, and keeps the water level within the valid range from zero to the tank capacity. It also identifies the first day on which the water level reaches or exceeds a specified threshold.

2. Rules and Conditions

Item	Condition / Action
Tank capacity	The maximum amount of water the tank can hold.
Initial water level	The water present in the tank before the simulation starts.
Daily inflow	Water added to the tank during a day.
Daily outflow	Water removed from the tank during a day.
Overflow	If the calculated level exceeds capacity, the extra water is lost and the level is set to capacity.
Shortage	If the calculated level becomes negative, the shortage is reported and the level is set to zero.
Threshold	Record the first day when the water level is greater than or equal to the threshold.
Final result	Display the final water level and the first threshold day, or state that it was never reached.

3. Algorithm

1. Start the program.

2. Read the tank capacity, initial water level, threshold level, and number of days.

3. Initialize first_threshold_day to -1 to indicate that the threshold has not yet been reached.

4. Repeat the following steps for each day from 1 to the given number of days.

5. Read the daily inflow and outflow amounts.

6. Calculate the new water level by adding inflow and subtracting outflow.

7. If the new level is greater than capacity, calculate the overflow, report it, and set the level to capacity.

8. Otherwise, if the new level is below zero, calculate the shortage, report it, and set the level to zero.

9. Display the current water level.

10. If the threshold has not been reached before and the current level is greater than or equal to the threshold, store the current day.

11. After all days, display the final water level and the first threshold day, if reached.

12. Stop the program.

4. Pseudocode

```
BEGIN
    INPUT capacity, current_level, threshold, days
    SET first_threshold_day = -1

    FOR day = 1 TO days
        INPUT inflow, outflow
        current_level = current_level + inflow - outflow

        IF current_level > capacity THEN
            overflow = current_level - capacity
            current_level = capacity
            DISPLAY overflow
        ELSE IF current_level < 0 THEN
            shortage = ABS(current_level)
            current_level = 0
            DISPLAY shortage
        END IF

        DISPLAY current_level

        IF first_threshold_day = -1 AND current_level >= threshold THEN
            first_threshold_day = day
        END IF
    END FOR

    DISPLAY final current_level
    IF first_threshold_day != -1 THEN
        DISPLAY first_threshold_day
    ELSE
        DISPLAY "Threshold was never reached"
    END IF
END
```

5. Concepts Used

Variables: capacity, current_level, threshold, days, inflow, outflow, overflow, shortage, first_threshold_day, and day. Data types: float, integer, and string. Operators: addition (+), subtraction (-), comparison (>, <, >=), and logical AND. Conditional statements: if and elif. Loop: for loop with range() to process each day. Built-in function: abs() to calculate the positive shortage amount. Formatted strings (f-strings) display values to two decimal places.

6. Conclusion

The Multi-Day Water Tank Simulation tracks the amount of water in a tank over several days. It handles overflow and shortage conditions, displays the water level after each day, and identifies when the threshold is first reached. This program demonstrates the use of variables, data types, operators, loops, conditional statements, and formatted output in Python.